

FTRIA-005 — Local and Global Degrees of Freedom

Part I — Foundations of Runtime-Invariant Degrees of Freedom

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Repository: <https://github.com/sizhet/Function-Tunnel-and-Runtime-Invariant-Algebra-FTRIA>

Abstract

Runtime Invariants rarely possess a single, uniform Degree of Freedom. Instead, different components of a Runtime Invariant often exhibit different levels of flexibility.

Some variations remain confined to local regions of the runtime structure, while others affect the Runtime Invariant globally.

This paper distinguishes between **Local Degrees of Freedom (LDoF)** and **Global Degrees of Freedom (GDoF)**.

Understanding this distinction enables more precise reasoning about runtime evolution, optimization, migration, and structural stability.

Rather than viewing Runtime Invariants as monolithic objects, FTRIA models them as hierarchically organized systems with multiple interacting levels of admissible variation.

1. Degrees of Freedom Are Hierarchical

The previous papers established that Runtime Invariants possess admissible regions bounded by structural constraints.

However,

these regions are rarely uniform.

Different parts of the Runtime Invariant permit different amounts of variation.

Consequently,

Degrees of Freedom naturally form a hierarchy.

Some variations remain local.

Others propagate throughout the entire Runtime Invariant.

2. Local Degrees of Freedom

A **Local Degree of Freedom (LDoF)** affects only a limited portion of the Runtime Invariant.

Typical examples include

- function refactoring,
- local optimization,
- cache tuning,
- instruction scheduling,
- module replacement,
- internal data structure optimization.

These modifications improve or reorganize local implementations while preserving both local functionality and global Runtime Invariant identity.

Conceptually,

Runtime Invariant

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| ○○○ |

| ○ LDoF ○ |

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Only a localized region evolves.

The overall Runtime Invariant remains stable.

3. Global Degrees of Freedom

A **Global Degree of Freedom (GDoF)** affects the Runtime Invariant as a whole.

Examples include

- distributed deployment,
- hardware migration,
- large-scale scheduling,
- runtime packaging,
- architecture-level optimization,
- execution model transformation.

Although the implementation changes globally,

the Runtime Invariant continues to preserve its structural identity.

Conceptually,

Entire Runtime Invariant

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Entire configuration changes

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The transformation spans the entire runtime system.

4. Local Freedom Supports Global Stability

One remarkable property of many Runtime Invariants is that extensive local variation can coexist with remarkable global stability.

For example,

individual modules,

functions,

or execution schedules

may evolve independently,

while the Runtime Invariant itself remains unchanged.

This balance between local flexibility and global stability is a defining characteristic of robust computational systems.

5. Coupling Between Local and Global Degrees of Freedom

Local and global Degrees of Freedom are not independent.

A sufficiently large accumulation of local changes may eventually influence global behavior.

Likewise,

global transformations often require coordinated local adaptations.

Therefore,

Runtime evolution should be viewed as interactions between multiple levels of admissible variation.

This interaction forms a hierarchical structure rather than a flat collection of transformations.

6. Structural Stability Across Multiple Scales

Runtime Invariants therefore possess stability at multiple structural scales.

Examples include

- instruction level,

- function level,
- module level,
- Calling Graph level,
- Function Tunnel level,
- Runtime Architecture level.

Each level possesses its own Degrees of Freedom and its own Constraint Surface.

Collectively,
these layers produce hierarchical structural stability.

7. Engineering Implications

Distinguishing between Local and Global Degrees of Freedom enables more effective runtime engineering.

Examples include

- localized optimization without global redesign,
- modular runtime upgrades,
- scalable distributed deployment,
- incremental runtime evolution,
- hierarchical verification,
- partial Runtime migration.

Rather than treating every modification as a complete system redesign, engineers may operate at the appropriate structural level.

This significantly improves scalability and maintainability.

8. Toward Hierarchical Runtime Geometry

The distinction between Local and Global Degrees of Freedom suggests that Runtime Geometry is inherently hierarchical.

Instead of one admissible region,

a Runtime Invariant contains nested admissible regions corresponding to different structural levels.

Future Runtime Geometry will therefore study

- nested Constraint Surfaces,
- hierarchical admissible regions,
- multi-scale Runtime evolution,
- coupled Runtime dynamics.

This perspective extends Runtime Invariants from static structural objects into hierarchically organized runtime spaces.

Key Takeaways

- Degrees of Freedom naturally exist at multiple structural scales.

- Local Degrees of Freedom affect only limited regions of a Runtime Invariant.
- Global Degrees of Freedom influence the Runtime Invariant as a whole.
- Extensive local variation may coexist with stable global Runtime Invariants.
- Local and global Degrees of Freedom interact through hierarchical structural coupling.
- Runtime Geometry is fundamentally hierarchical rather than uniform.

Related FTRIA Documents

- **FTRIA-001 — Runtime Invariant Degrees of Freedom**
- **FTRIA-002 — RI-Preserving Transformation**
- **FTRIA-003 — Admissible Runtime Variation**
- **FTRIA-004 — Constraint Surfaces and Invariant Boundaries**
- **FTRIA-006 — Coupled Degrees of Freedom**